

2-9 Use Addition to Subtract

Learning goal: I can use a bar diagram and a related addition equation to solve a subtraction problem.

Textbook: pages 67–70. The book shows $575 - 246 = ?$ and $246 + ? = 575$ with one bar diagram.

Words to Know

bar diagram	A drawing with bars that shows the whole and its parts.
whole / part	The whole is the total amount. The parts are the amounts that make up the whole.
unknown	The number we do not know yet. We write it as ?.
addend	A number you add. In $248 + ? = 563$, the ? is an unknown addend.
related equations	Equations that use the same three numbers, like $563 - 248 = 315$ and $248 + 315 = 563$.
count up	Start at the part and add jumps until you reach the whole.

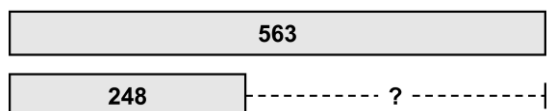
How to Teach It

1. Draw a bar diagram: the **whole** is the long bar on top. The known **part** is below it. The dashed piece with ? is the part you need.
2. Write the subtraction: whole – part = ?. Then write the related addition: part + ? = whole.
3. Solve the addition by **counting up** from the part: jump to the next ten, then to the next hundred, then to the whole. Add all the jumps.
4. Check: part + answer should equal the whole.
5. Watch for adding the two numbers (the answer would be bigger than the whole). Ask, “Can the answer be bigger than the whole?”

Tip: Ask, “Which number is the whole? What do you add to the part to get the whole?”

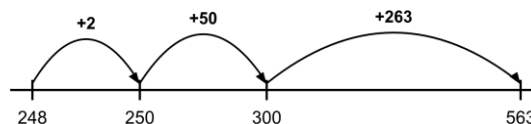
Worked Examples

Example 1: Bar diagram and related equations



$563 - 248 = ?$ is the same as $248 + ? = 563$

Example 2: Count up to solve



$2 + 50 + 263 = 315$. Check: $248 + 315 = 563$ ✓

Example 3: Subtract across zeros

$400 - 153 = ? \rightarrow 153 + ? = 400$

Count up: $153 \rightarrow 160 (+7) \rightarrow 200 (+40) \rightarrow 400 (+200)$.

$7 + 40 + 200 = 247$

Example 4: Watch for these mistakes

To find $458 - 302$, do **not** add $302 + 458$. Solve $302 + ? = 458 \rightarrow 156$.

$452 - ? = 237$: the missing number is a part. Solve $237 + ? = 452 \rightarrow 215$.

Skills Checklist Check off each skill your child can do without help.

- | | |
|---|---|
| ☐ Fill in a bar diagram for a story. | ☐ Count up across zeros (like $400 - 153$). |
| ☐ Write a related addition equation for a subtraction. | ☐ Check a difference with addition. |
| ☐ Count up to find an unknown addend. | ☐ Explain why adding the two numbers is wrong. |

2-9 Use Addition to Subtract

Name _____ Date _____

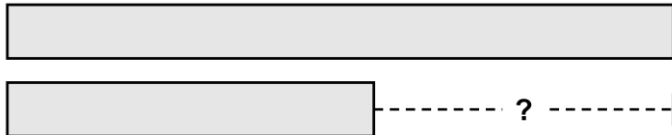
Together *Work with your grown-up. Talk about each step.*

- A.** Write a related addition equation for $584 - 342 = ?$. Then count up to solve.

Addition equation: _____

Answer: _____

- B.** A book has 500 pages. Ben has read 263 pages. How many pages are left? Label the bar diagram. Then write an addition equation and solve.



Equation: _____

Pages left: _____

- C.** Lila wants to use addition to solve $728 - 315 = ?$. How can you explain her strategy? What is the answer?

Answer: _____

On My Own *Try these by yourself. Show your work.*

- 1.** Write a related addition equation. Then solve.

$$847 - 413 = ?$$

Addition equation: _____

Answer: _____

- 2.** Write a related addition equation. Then solve.

$$300 - 162 = ?$$

Addition equation: _____

Answer: _____

3. Write a related addition equation.
Then solve.
 $936 - 255 = ?$

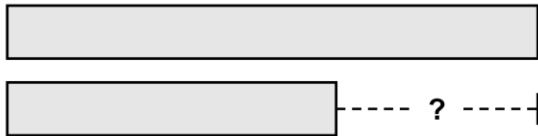
Addition equation: _____

Answer: _____

4. Omar adds $205 + 638$ to find $638 - 205$. How can you help him use addition the right way?

5. On Saturday, 468 people visited the zoo. On Sunday, 735 people visited. How many more people visited on Sunday?

a. Label the bar diagram.



b. Write an equation with an unknown.
Solve.

Equation: _____

Answer: _____

6. Use addition to find the difference.
 $590 - 347 =$ _____

7. Use addition to find the difference.
_____ $= 600 - 274$

8. Use addition to find the difference.
 $804 - 368 =$ _____

9. Use addition to find the difference.
 $321 - 156 =$ _____

10. Mia and Jo picked 400 apples in all.
Mia picked 186 apples. How many
apples did Jo pick?

11. A club has \$900 for a trip. The bus
costs \$638. How much money is left?

Apples: _____

Money left: _____

12. Ray says he can solve $563 - ? = 327$ by finding $563 + 327$. Do you agree? Explain.

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Together

A	$342 + ? = 584$. Count up: $342 \rightarrow 384 (+42) \rightarrow 584 (+200)$. Answer: 242 <i>Also accept $? + 342 = 584$. Taught route: $342 \rightarrow 350 (+8) \rightarrow 400 (+50) \rightarrow 584 (+184) = 242$. Any correct jumps are fine.</i>
B	Whole bar: 500 . Part bar: 263 . ?: pages left. $263 + ? = 500$. $263 \rightarrow 270 (+7) \rightarrow 300 (+30) \rightarrow 500 (+200)$. 237 pages
C	Think $315 + ? = 728$: what do you add to 315 to get 728? Count up: $315 \rightarrow 328 (+13) \rightarrow 728 (+400)$. Answer: 413 <i>Taught route: $315 \rightarrow 320 (+5) \rightarrow 400 (+80) \rightarrow 728 (+328) = 413$. Accept any explanation that turns the subtraction into "part + ? = whole."</i>

On My Own

1	$413 + ? = 847 \rightarrow 434$ <i>Items 1–3: also accept $? + 413 = 847$, etc.</i>
2	$162 + ? = 300 \rightarrow 138$ $162 \rightarrow 170 (+8) \rightarrow 200 (+30) \rightarrow 300 (+100)$.
3	$255 + ? = 936 \rightarrow 681$
4	Adding makes a number bigger than 638, the whole. He should solve $205 + ? = 638$. Answer: 433 .
5	a. Whole (long bar): 735 . Part: 468 . ?: how many more. b. $468 + ? = 735$ (or $735 - 468 = ?$). Answer: 267
6	243 ($347 + 243 = 590$) <i>Items 6–9: check each answer by adding it to the smaller number.</i>
7	326 ($274 + 326 = 600$)
8	436 ($368 + 436 = 804$)
9	165 ($156 + 165 = 321$)
10	$186 + ? = 400 \rightarrow 214$ apples
11	$638 + ? = 900 \rightarrow \262
12	No. 563 is the whole, so the missing number is a part: $327 + ? = 563$. Answer: 236 . $563 + 327 = 890$ is bigger than the whole.